

DNA Double Helix and Base Pairing

Introductory biology · ~2 pages · for XR EduAgent study

Learning goals

You should be able to: (1) name the four bases; (2) state complementary pairing rules; (3) explain why pairing is specific; (4) relate base pairs to the double-helix structure.

Structure snapshot

DNA is a double helix: two strands of nucleotides wind around a common axis. Each nucleotide has a sugar, a phosphate, and a nitrogenous base. The backbone is sugar–phosphate; bases point inward.

The four bases

Adenine (A), Thymine (T), Guanine (G), and Cytosine (C). In RNA, uracil (U) replaces thymine.

Complementary pairing

A pairs with T (two hydrogen bonds). G pairs with C (three hydrogen bonds). This complementarity lets each strand act as a template for the other during replication.

Antiparallel strands

The two strands run in opposite directions ($5' \rightarrow 3'$ vs $3' \rightarrow 5'$). Base pairs stack like rungs of a ladder; the helix turn and major/minor grooves matter for protein binding, but pairing rules are the core idea here.

Why spatial models help

A 3D helix with clickable bases makes A–T vs G–C pairing and antiparallel orientation easier to remember than a flat textbook diagram alone.

Study use only · Short excerpt prepared for XR EduAgent usability study · Not a full textbook chapter.